

The Effects of Organizational Structure on Communication Within Audit Teams

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SUMMARY

Some audit researchers suggest that high levels of audit team structure may encumber communication within audit teams by impeding information-gathering activities. Others suggest that structure benefits communication by coordinating and controlling information flows. This study evaluated these arguments by examining the relationship between the structure of audit teams and selected communication variables (information overload, boundary spanning, satisfaction with supervision, and accuracy of information). Questionnaire data were gathered from a national sample of 109 audit teams, with three auditors responding from each team (i.e., $n = 327$). Information overload, satisfaction with supervision, and accuracy of information were less in audit teams with greater structure. The implication is that the level of structure adopted by teams has both positive and negative effects on communication, with structured teams providing greater control over information overload but impairing satisfaction with supervision and the accuracy of information.

Key Words: Organizational structure, Audit structure, Communication, Audit teams.

Data Availability: Please contact the first author concerning data availability.

INTRODUCTION

An audit team is a planned hierarchical assemblage of individuals (e.g., seniors, supervisors and managers) brought together for the purpose of conducting an audit (cf., Solomon 1987). It is the basic operating unit that performs audit work in accounting firms. Because audit activities are distributed among members of the team, a key determinant of team success is the degree with which individual efforts coalesce into a well-informed audit opinion. Communication among team members is the fundamental activity through which discrete efforts are transferred for integration into the end product, the audit opinion. Successful coalescence depends on informative communication within the team.

The quantity and quality of information exchanged within an audit team may be contingent

upon the organizational structure of the team (hereafter denoted as "audit team structure"). In line with the more general term, organizational structure (e.g., Burns and Stalker 1961), audit team structure refers to the degree with which the team has imposed constraints on the activities and behaviors of its members. The purpose of imposing constraints is to assure the proper level of coordination and integration of team activities to achieve audit goals and objectives. General approaches that may be adopted to structure the team include: (1) standardization and programming of team activities (e.g., using

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pre-established rules, routines, blueprints, or schedules; Dirsmith and McAllister 1982b),¹ (2) concentration of decision-making authority in higher organizational levels of the team (Bamber and Bylinski 1982), and (3) restrictions on the ability of team members to make changes as needed (Dirsmith and McAllister 1982b). The constraints that such approaches place on the behavior of individual team members have a bearing on the communication within the team (e.g., Klaus and Bass 1982). Systemized audit approaches constrain the kinds and amounts of information that are gathered and transmitted to team members; concentration of decision making in higher organizational levels suggests that senior team members will have different informational needs than counterparts of less structured teams; and restricted ability to make changes suggests that auditors will need to consult with superiors more often.

While audit team structure may govern both the frequencies with which team members interact to exchange information and the kinds and amounts of information exchanged when they do interact, researchers appear to have reached different conclusions as to its advantages with regard to communication. For instance, Cushing and Loebbecke (1986, 42) present a possible advantage. They argue that standardization and programming of audit approaches may provide auditors with a common ground for information exchanges within the team. Bamber and Bylinski (1982, 41) propose a disadvantage. They argue that a cost of increased formalization and standardization associated with structured environments is "reduced information processing capacity" of the team.

The purpose of the present study is to determine whether audit team structure is associated with four criterion variables that reflect the quantity and quality of information exchanged within the audit team. The four criterion variables are information overload, boundary spanning, satisfaction with supervision, and accuracy of information. Information overload is the exchange of greater quantities of information than auditors can efficiently use. It has been shown to reduce the effectiveness of decision making (Snowball 1980). Boundary spanning is the de-

gree with which team members communicate with people outside of the team, such as statistical specialists, for the purpose of acquiring audit-related information. The inflow of relevant information from external sources should augment the quantity and quality of information exchanged within the team. Satisfaction with supervision is the degree with which audit team members receive satisfactory guidance from their supervisors. Supervision suggests downward communication in the form of direction and advice about task-related matters. Satisfaction with supervision reflects the quantity and quality of the information conveyed during such exchanges. Accuracy of information assesses the quality of information exchanged within the team (e.g., Bamber et al. 1989).

HYPOTHESES

Audit team structure is represented in terms of Burns and Stalker's (1961) framework of organizational structure. To apply the framework, organizations or organizational subunits (e.g., teams) are typed as either mechanistic or organic on the basis of their organizational characteristics. Broadly, the characteristics relate to systematic, standardized and formalized approaches adopted for the execution of their respective functions (e.g., Birnbaum and Wong 1985) and to the centralization of decision making in higher levels of the organization (e.g., Miller and Droge 1986).

Mechanistic forms of organization, which are described with adjectives such as formalized, standardized, centralized, authoritarian and

¹ Systematic approaches for conducting audits (e.g., Cushing and Loebbecke 1986; Kinney 1986; Bamber et al. 1993) would fall within this dimension of organizational structure. However, the focal point of these studies on audit structure may be incommensurate with this study's focus on team structure. As Cushing and Loebbecke (1986, 32) note, "the concept of the mechanistic audit should not be equated with our description of highly structured audit approaches, because (despite their similarities) there are significant differences between the two concepts." A primary difference, for instance, is that team structure is also concerned with the degree to which formal authority is concentrated in individuals usually high in the organization, "thus permitting employees (usually low in the organization) minimum input into their work" (Robbins 1983, 78).

nonparticipatory, impose many constraints on decision makers and are appropriate for organizations that operate in stable environments. Conversely, organic forms of organization, which are described with adjectives such as self-managing, autonomous, informal, self-actualizing and participative, place fewer constraints on decision makers and are appropriate for organizations that must deal with uncertain environments (see Courtright et al. 1989). Burns and Stalker's (1961) framework has been applied to public accounting organizations (Watson 1975; Zanzi 1987) and to teams (Bamber and Bylinski 1982, 41–42). Bamber and Bylinski (1982) opine that, in general, audit teams are organized with mechanistic structures.

One of the proposed consequences of mechanistic structures is that they impose constraints on communication (e.g., Klauss and Bass 1982, 1–34). These constraints form the basis for hypothesizing correlational relationships between audit team structure and the four criterion communication variables in this study.

Information Overload

The boundaries that mechanistic structures place on the behavior of team members may help the team attain a level of coordination and control that is appropriate for the risk characteristics of the audit client (e.g., low inherent risk) and the makeup of the team (e.g., low levels of audit experience). With regard to communication, for instance, audit teams may opt for greater structure (e.g., formal channels of communication and standardized reports) as a way to cope with the problem of information overload (cf., Hall 1996, 183–185). Work boundaries in mechanistic structures are more defined, and information gathering and processing are more mechanical and better delineated, which has the effect of controlling the quantity of information that is exchanged among decision makers (e.g., Bacharach and Aiken 1977). Accordingly, we propose the following:

- H1:** There is a negative correlation between information overload and audit team structure.

Boundary Spanning

A group of activities that may bear on whether team members exchange appropriate sets

of information for decision making is boundary spanning. It occurs when auditors span the bounds of the audit team to access others for information (e.g., Danos et al. 1991, 107). Examples of boundary-spanning activities include communication across team boundaries with members of other teams or specialists in order to obtain information and reduce uncertainty in the audit environment (e.g., Leifer and Delbecq 1978). As a communication measure, boundary spanning represents the frequency with which the team enters into informational exchanges with individuals outside the confines of the team. Audit-related information acquired through boundary spanning should serve to enhance communication that takes place within the hierarchical structure of the team. Inflows of information into the team as a result of boundary spanning increase the quantity of information that is available for exchange within the team. In addition, spanning the bounds of the team to acquire information from specialists expands the capacity of team members to make more precise, accurate judgments than might otherwise be made. Upgrading individual judgments through boundary spanning elevates the quality of information that is available for exchange within the team.

Leifer and Huber (1977) found that mechanistic structures are associated with reduced levels of boundary spanning as an information-gathering activity. They conclude that mechanistic structures impose constraints on the level of boundary spanning that takes place in the organization. In contrast to Leifer and Huber (1977), auditing studies have found what appears, on the surface, to be an opposite relationship, that auditors of firms with more structured audit approaches are more likely to span the bounds of the team for advice. For instance, Bamber and Snowball (1988) and Prawitt (1995) had auditors of four firms, two with structured audit approaches and two with unstructured audit approaches, indicate the likelihood that they would consult with specialists. Bamber and Snowball (1988, 502, table 4) found that one structured firm was more likely to consult experts than each of the others. Prawitt (1995, 460) found that audit managers of structured firms have a greater

likelihood of using specialists in higher complexity environments. It may be that, as some argue, structured audit approaches imposed at the firm level promote a decline in the quantity and quality of information exchanges within audit teams. To make up for the decline, the organizational structure of audit teams may have been relaxed in order to impose fewer constraints on boundary spanning. In contrast to Bamber and Snowball (1988) and Prawitt (1995), Danos et al. (1991, 107) failed to find support for a relationship between structured audit approaches and boundary spanning. They studied firms that were classified by Kinney (1986) as "intermediate" and "structured" and found that firms of either type do not have extensive consultation outside of the audit team.

Because we are studying organizational structure of audit teams, rather than structured audit approaches, we advance a hypothesis that is in line with Leifer and Huber's (1977) argument that mechanistic structures are more likely to constrain boundary spanning. Thus, we propose the following:

- H2:** There is a negative correlation between boundary spanning and audit team structure.

Satisfaction With Supervision

Another indication of the level of communication within audit teams is the quality and quantity of information auditors receive from supervisors. Supervision entails a process of directing the activities of subordinates. It includes downward communication regarding task instruction, task objectives, helpful suggestions, and constructive assessments of preliminary plans and past decision making (e.g., Hall 1996, 174–175). If the downward communication of these kinds of information is deficient, auditors may be ill-equipped to resolve important issues concerning how to perform tasks and interpret audit information. The quantity and quality of information received from supervisors was measured in terms of recipients' satisfaction with supervision, an attitude that reflects the degree with which perceived informational needs are fulfilled. Informational needs are fulfilled more completely, for instance, if a sufficient amount of information

of sufficient quality is received from supervisors.

The organizational literature (e.g., Mintzberg 1983) suggests that the level of accessibility subordinates have to supervisors for counseling about task-related matters may be less in mechanistic structures. The rigidity and specificity that mechanistic structures impose on the performance of team duties neither support nor encourage counseling activities by superiors. In mechanistic structures, standardization supplants supervision as the control process (Mintzberg 1983, 4–5). Two-way, superior-subordinate communication is restricted in favor of a more top-down orientation to the direction and initiation of the flow of information (e.g., Courtright et al. 1989). Without access to superiors, auditors may not have the opportunity to collect needed information, such as clarifications and how-to assistance, from more senior members of the audit team.

Bamber and Snowball (1988, table 4) found that auditors of firms with structured audit approaches were less likely to consult with an engagement manager, which suggests that they seek guidance less frequently. Prawitt (1995) presents findings that suggest a reason why they may be reluctant to consult with an engagement manager. He found that audit managers of firms with structured audit approaches tend to assign less-experienced auditors to supervisory roles. The advice that these individuals can provide to subordinate team members may be viewed as wanting. Thus, we propose the following:

- H3:** There is a negative correlation between satisfaction with supervision and audit team structure.

Accuracy of Information

The variable, accuracy of information, reflects the general quality of the information being exchanged within a team. A potential drawback of mechanistic structures is that they foster rigid adherence to rules, which may impair the quality of information exchanged with decision makers. High degrees of formalization of the audit process may encourage team members to follow prescribed rules "for the sake of the rules themselves," since compliance with rules forms the basis of performance evaluations (Hall

1996, 70). Rules may become more important to team members than the goals that the rules were designed to accomplish. Cushing and Loebbecke (1986, 43) label this mind-set as "mechanistic thinking." Failures of individual team members "to observe important facts, or...to reason through to appropriate judgments and conclusions" (Cushing and Loebbecke 1986, 43) may impoverish the quality of information exchanged within the team. Over time, the flow of deficient information may infuse doubt as to the accuracy of information being exchanged within the team.

Cushing and Loebbecke (1986, 42) suggest a benefit of structure that may enhance a perception of information accuracy. Audit structure may provide "a common terminology with which all members of the audit team are familiar....This may...reduce the possibility of misunderstandings with respect to the requirements of each audit job." There is some support for this view. Bamber et al. (1989, table 2) found that audit seniors' perception of communication adequacy (accuracy and timeliness) was greater in structured firms than in unstructured firms. However, the difference achieved only a 10 percent level of statistical significance with a one-tailed test. This counter finding aside, we propose the following:

- H4:** There is a negative correlation between accuracy of information and audit team structure.

SAMPLE AND PROCEDURES

The initial sample included 448 audit teams. The sample was drawn from two sources. First, 380 offices of local, regional and national firms were selected from those listed in *Emerson's Directory of Leading U.S. Accounting Firms* (Emerson 1992).² To be eligible for selection, an office had to have a staff of 25 or more professionals in order to assure that offices were suitable in size to have reasonably large audit teams. Each office was requested to provide one team. Second, representatives of two national firms identified 68 audit teams operating within 13 offices. Each

office employed 50 or more professionals and was located in a large metropolitan area.

The managing office partner or his or her designee was instructed to distribute questionnaires to three auditors of each audit team. The selected audit team had to be currently engaged in an audit or to have recently completed an audit engagement. This team-selection rule helped to assure that members of the team could unambiguously bring the audit engagement to mind. Additionally, each participating office was asked to choose, to the extent possible, audit team members who had 3–5 years of auditing experience. This criterion was added because auditors with more experience are generally involved with more than one audit engagement at a time and auditors with less experience generally do not have in-charge responsibility. However, auditors with experience levels outside of these boundaries were included for study if they were working primarily on one audit team (75 percent of total chargeable time in an average week) or had in-charge responsibility. For an audit team to be included in the study, all three team members had to return completed questionnaires.

Offices representing 226 teams (50 percent) responded to the request for data. However, 79 offices indicated that they did not have an audit team that met the selection criteria. An additional 34 teams were excluded due to nonresponse of at least one team member and four teams were excluded due to incomplete responses of one or more team members. Thus, the final sample is made up of 109 teams (24 percent).³

² The sample of teams is not randomly selected and therefore does not generalize to offices listed in Emerson (1992) due to an over representation of larger offices.

³ The actual response rate is indeterminable because many offices included in the initial sample did not meet sample criteria. Of 226 teams that responded to the request for data, 35 percent indicated in their response that their office did not meet sample criteria. When applied to the initial sample, this percentage indicates that 291 teams out of the 448 sampled met sample criteria. Using 291 teams as the base, the 109 responding teams represent a response rate of 37 percent.

Characteristics of the final sample are reported in table 1.⁴

Each participating auditor completed a 70-item questionnaire. Packets containing survey material were sent to managing office partners of each local firm, regional firm and national office without a firm or office coordinator. Accompanying the packets were instructions to managing office partners outlining the selection criteria that were to be used in choosing audit teams and audit team members for the study and explaining how to handle distribution of the survey material to team members. For the two national firms with firm coordinators, packets were mailed directly to the office contact person selected by the co-

ordinator. The office contact person then selected the audit teams and team participants according to the selection criteria and handled distribution of the survey material to each participant. Four weeks after mailing the survey

⁴ Two of the characteristics in table 1 (office size and geographic location) were used to test the sample for nonresponse bias. This testing excluded the 68 teams identified by the two national firms. The frequency distribution for the initial sample was not significantly different from the distribution for responding offices ($X^2(3)=7.38$, n.s., for office size and $X^2(3)=1.72$, n.s., for geographic location) or from the distribution of offices in the final sample ($X^2(3)=4.03$, n.s., for office size and $X^2(3)=6.11$, n.s., for geographic location).

TABLE 1
Characteristics of the Sample

Office size:	
25–50 professionals	11 teams (10%)
51–75 professionals	23 teams (21%)
76–100 professionals	14 teams (13%)
over 100 professionals	61 teams (56%)
Geographic location of office:^a	
Northeast	22 teams (20%)
North central	36 teams (33%)
South	37 teams (34%)
West	14 teams (13%)
Number of cities represented	60 cities
Number of states represented	30 states
Firm size:	
National firms (7 firms)	73 teams (67%)
Regional firms	15 teams (14%)
Local firms	21 teams (19%)
Team members' mean age	27.5 years (sd = 4.4)
Team members' gender:	
Female	139 auditors (43%)
Male	188 auditors (57%)
Respondents' position:	
Staff auditor	60 auditors (18%)
Senior auditor	177 auditors (54%)
Manager	84 auditors (26%)
Partner	6 auditors (2%)
Average team size	7.1 auditors (sd = 4.7)
Team members' mean level of auditing experience	4.6 years (sd = 3.4)

^a Based on the 1980 Census classification of northeast, south, north central, and west geographic regions (U.S. Department of Commerce 1984).

instrument, follow-up letters were mailed to managing office partners (or office contact persons) of offices which had not responded. A second follow-up attempt was made with nonresponding offices four weeks after the first attempt. Completed instruments were mailed directly to the researchers.

MEASURES

The criterion variables (information overload, boundary spanning, satisfaction with supervision, and accuracy of information) and the independent variable (audit team structure) were measured with multiple-item scales. The appendix contains a list of the items included in each scale.

Audit Team Structure

Audit team structure was measured with Leifer's (1975) 21-item mechanistic-organic scale, in which characteristics of organizational structure are rated on agree-disagree response scales. Leifer's scale has been used previously to measure the mechanistic structure of work groups (Leifer and Huber 1977). Organizational characteristics represented include: (1) centralization of, and participation in, decision making, (2) distribution of authority and control, (3) formalization, and (4) standardization. Leifer and Huber (1977) report acceptable Spearman-Brown reliabilities for the 21-item scale, and report that the scale has adequate construct validity. Individual items were adapted to fit the context of an audit team. Individual responses were made on seven-point response scales and were coded such that higher scores indicate greater (more mechanistic) audit team structure.

Information Overload

Information overload assesses the frequency (very often to not often) and extent (not much to too much) with which team members exchange too little or too much information. It was measured with a three-item scale used previously in studies by Roberts and O'Reilly (1974) and O'Reilly (1980). The scale was modified to fit an audit context. An example item from this scale is, "In a typical week on the audit team, approximately how often do you have less than the

amount of information you can consistently handle for making the best possible work-related decision?" O'Reilly (1980) reports Cronbach's alphas of .74 and .71 for samples of managers and navy aviators, respectively, and concludes that the scale has adequate construct validity. Individual responses were made on a seven-point response scale, with higher scores representing greater information overload.

Boundary Spanning

Boundary spanning is measured in terms of the number of interactions that team members have in an average week with people: (1) outside of the audit team but within the office and (2) outside of the office altogether. The specific questions posed to assess the number of interactions of each category were adapted from questions in Leifer's (1975) boundary-spanning scale.⁵ An example question includes: "How many hours (in an average week) do you attend formal meetings in regard to your assigned work on the audit team?" Auditors provided ballpark estimates to each question. Higher numbers on the scale indicate greater boundary spanning.

Satisfaction With Supervision

The satisfaction-with-supervision scale contained two items, which were adapted from two items included in an instrument developed by Downs and Hazen (1977). The two items are related to the amount of supervision given on the team and the guidance provided by supervisors on the team in solving work-related problems. They were shown by Gregson (1990, 40) to load on the same factor. Individual responses were made on a seven-point response scale, with higher scores indicating greater satisfaction with supervision.

Accuracy of Information

Accuracy of information was measured with a five-item scale previously used by O'Reilly and Roberts (1976, 1977). The scale measures the

⁵ Leifer and Huber (1977) examine the construct validity of the questions in the Leifer (1975) scale and report that it was adequate for a sample of 222 employees in a large health and welfare organization.

perceived accuracy of information exchanged among members of the team during the audit engagement. O'Reilly and Roberts (1976, 1977) report internal consistency reliability coefficients (Cronbach's alphas) of .74, .73, .75 and .78 for four different samples, which suggests that the scale items are reasonably homogeneous. The items were modified to fit an audit context. Individual responses were made on a agree-disagree, seven-point response scale, with higher scores on the scale indicating greater accuracy of information.

RESULTS

The proposed relationships were tested with correlational techniques (LISREL 8 computer software; Joreskog and Sorbom 1993). The unit of analysis for the study is the audit team. A team score for each variable was computed by summing the individual item responses provided by the three team members. For the variables under study, the proportion of total variance that is shared among teammates (Spearman-Brown formula; DeVellis 1991) is .26 for information overload, .42 for boundary spanning, .42 for satisfaction with supervision, .22 for accuracy of information, and .15 for audit team structure.

The length of the instrument used to measure audit team structure severely restricted the degrees of freedom available to perform a LISREL analysis of the data. To preserve sufficient degrees of freedom to perform a LISREL analysis, we formed three indicators of structure out of the set of 21 scale items. Following a procedure used by Mathieu (1991), the 21 items were separated into three subgroups of seven items each. In the first round, the two items with the highest and lowest loading were placed in group 1. In the second round, the two items with the next highest and next lowest loading were placed in group 2. Finally, in the third round, the two items with the next highest and lowest loading were placed in group 3. The three items leftover were assigned randomly to the three groups, resulting in three subgroups of seven items each with Cronbach's alphas of .46, .52 and .63, respectively. Subgroups formed in this manner provide parallel indicators of structure that preserve degrees of freedom without forfeiting the measurement of

scale reliabilities needed for LISREL to adjust structural path estimates for measurement error (e.g., Rahim and Magner 1995).

Measurement Properties of the Scales

The measurement properties of the multiple-item scales were evaluated to assure that scales were acceptable measures for use in hypothesis testing. Four tests were performed using confirmatory factor analysis with the LISREL 8 computer package.⁶

Three tests were performed to evaluate the convergent validity of the multiple-item scales. For scales with high convergent validity, scale items should be highly correlated with one another due to the similarity they each have with the attribute being measured by the scale. These tests utilize LISREL-generated estimates of standardized item loadings, which are indicators of the degree of association between each scale item and its respective latent variable. First, the standardized item loadings are significant for each item. Second, composite reliability, which is analogous to Cronbach's alpha and provides a global measure of the intercorrelation among the individual items of a scale, are all above the .70 threshold for adequate internal reliability recommended by Nunnally (1978, 245).⁷ Third, variance extracted estimates, which represent the proportion of variance in the data that is captured by a latent variable (see Fornell and Larcker 1981), range from marginally acceptable (e.g., .49 for accuracy of information) to acceptable (e.g., .74 for satisfaction with supervision). These

⁶ The LISREL analysis involved the testing of a measurement model in which each of the 15 items making up the individual scales is allowed to load only on its respective factor and the factors are allowed to correlate. The analysis was performed using a covariance matrix and the maximum likelihood method.

⁷ Cronbach's alphas are as follows:

Information overload	.78
Accuracy of information	.81
Structure of the team	
3-item (reconstructed) scale	.88
Full 21-item scale	.77

The pairwise correlation between the two items of the satisfaction-with-supervision scale is .40 ($p < .001$), and the pairwise correlation between the two items of the boundary-spanning scale is .64 ($p < .001$).

results support the convergent validity of the scales.

To determine whether the scales measure different constructs (discriminant validity), two scales were united into a single scale. If the two scales measure separate attributes, there should be a statistically significant reduction in explained variation in the data when they are combined and treated as a single variable. A significant reduction in explained variation was found for each of the possible pairings of variables. This result supports the discriminant validity of the scales.⁸

Hypothesis Testing

The hypotheses were tested by examining the direction and significance of LISREL-estimated correlation coefficients between audit team structure and the criterion communication variables.⁹ The correlations represent the standardized paths of a structural model that relates team structure causally with the four criterion variables. The principal advantage of LISREL-generated correlations over OLS-computed correlations is that the effects of measurement errors in the observed variables, which generally attenuate correlations between underlying latent constructs, are eliminated from the estimates. The estimated correlations, and indicators of the

overall fit of the estimated model, are reported in table 2.

Three indicators of fit were computed. The Chi-square fit statistic is significant (114.65, $df = 86$, $p < .05$), which indicates unsatisfactory fit of the overall model to the data. It is not unusual, however, to have a significant Chi-square statistic, even when the model fits the data reasonably well (Bentler and

⁸ Two additional tests of discriminant validity were performed. First, the correlation coefficients between pairs of variables are significantly below perfect correlation (see Bagozzi 1980; Anderson and Gerbing 1988). Second, the proportion of variance in the data explained by the latent variable (i.e., the variance-extracted estimate) is more than the amount of variation in the data that can be explained by the correlation between two latent variables.

⁹ The LISREL-estimated squared pairwise correlations between criterion variables are as follows (each non-significant):

Information Overload with Accuracy of Information	.10
Information Overload with Boundary Spanning	.03
Information Overload with Satisfaction with Supervision	.11
Accuracy of Information with Boundary Spanning	.05
Accuracy of Information with Satisfaction with Supervision	.18
Boundary Spanning with Satisfaction with Supervision	.06

TABLE 2
LISREL-Estimated Correlations of the Communication
Variables With Audit Team Structure
($n = 109$)

Audit Team Structure with:

Information Overload
Boundary Spanning
Satisfaction with Supervision
Accuracy of Information

Correlation Coefficient

-.25 ($t = 2.10$, $p < .05$)
-.13 ($t = 1.09$, n.s.)
-.45 ($t = 3.68$, $p < .001$)
-.40 ($t = 3.15$, $p < .001$)

Goodness-of-Fit Statistics for the LISREL model:

Chi-square ($df = 86$) = 114.65, $p < .05$
Relative Noncentrality Index (RNI) = .95
Root Mean Square Error of Approximation (RMSEA) = .056
90% Confidence Interval for RMSEA = .023 to .081

Bonett 1980). Two additional fit indicators that are less sensitive to sample size were computed: (1) the relative noncentrality index (RNI), which has been recommended as the best available measure of fit (Gerbing and Anderson 1993), and (2) Steiger's (1990) root mean square error of approximation (RMSEA). The general rule is that RNI above .90 (Bentler and Bonett 1980) and RMSEA below .08 (Browne and Cudeck 1993) indicate reasonable fit. The estimated values of RNI (.95) and RMSEA (.056) fall within these guidelines. Thus, we conclude that the model is sufficiently specified to warrant examination of estimated correlations.

The correlations between audit team structure and the criterion variables are significant for information overload ($-.25$), satisfaction with supervision ($-.45$), and accuracy of information ($-.40$). The negative direction of the correlations indicates support for H1, H3 and H4. Information overload tends to be less in teams with greater audit team structure. In addition, information exchanged within teams is perceived less accurate and there is less satisfaction with supervision in teams with greater audit team structure. The correlation between audit team structure and boundary spanning ($-.13$) is nonsignificant.

DISCUSSION

The literature has provided mixed signals as to whether mechanistic or organic audit team structures are more appropriate for audit teams. Watson (1975) concluded that mechanistic structures were better suited for audit teams than management service teams. Other accounting researchers (e.g., Dirsmith and McAllister 1982a, 1982b), arguing for fewer constraints in the audit environment, have advocated the flexibility and adaptability of an organic structure. Together, the findings of this study suggest the possibility that audit team structure may have both beneficial and detrimental consequences on the quantity and quality of information that is exchanged within the audit team.

On one hand, the data supported the hypothesis that audit team structure is negatively correlated with information overload. This finding may reflect a benefit of team structure, which

is to regulate the quantity of information exchanged within the team. Team structure may reduce the tendency to provide team members with more information than they can efficiently handle for decision making. On the other hand, the data supported two hypotheses that suggest team structure may have a detrimental effect on team communication. Team structure was correlated negatively with accuracy of information and satisfaction with supervision. These findings may reflect the tendency of highly structured teams to exchange lower quality information for decision making and to experience less satisfactory levels of supervision.

Future research may well show that the appropriate level of audit team structure is dependent upon factors relating to the risk characteristics of the client and to the expertise of auditors on the team. There exists evidence to suggest that a context dependency may relate to the degree of audit uncertainty confronting the team. A team's need for information is a function of task uncertainty (e.g., Tushman 1978; Bamber and Bylinski 1982). If the client's audit environment is simple and stable, rigid application of specific audit procedures, rules and steps may provide the advantages of coordination and control while generating sufficient relevant information with which to render an informed audit opinion (e.g., Danos et al. 1991). However, when audit uncertainty increases, researchers argue that more discretion should be given to team members to modify their activities and recommend the less constraining environment of organic structures to allow this freedom (e.g., Dirsmith and McAllister 1982b). The linkages between task uncertainty and organizational structure of the team should provide a useful base from which to search for the mitigating role of audit context in the relationship between the structure of audit teams and communication.

This study should be evaluated in terms of its limitations. First, justification of each hypothesis is drawn from arguments of organizational theorists who characterize structure in terms of constraints imposed on activities and behaviors of organizational members. Prawitt (1995) presents empirical results that suggest an alternative explanation as to why teams with greater

audit team structure may exhibit lower accuracy of information and lower satisfaction with supervision. He reasoned that structured audit approaches would allow firms to assign less experienced auditors to audit tasks. In support of the proposition, he found that the average experience levels of auditors assigned to audit tasks and to supervision/review duties were lower in firms with structured audit approaches. If reduced experience is a differentiating characteristic of teams with greater audit team structure, the lower levels of information accuracy and satisfaction with supervision associated with mechanistic teams may be manifestations of the level of expertise of team members.

A second limitation is that team structure and communication variables were measured with perceptual techniques that may have generated systematically biased measurements. For instance, elements of team structure (e.g., reduced participation in decision making) may foster negative sentiments toward team members, the team, or the firm. Negative sentiments may manifest as inflated ratings of structure and as more nega-

tive responses to questions posed to measure variables such as accuracy of information and satisfaction with supervision. Therefore, the negative relationship between team structure and variables such as accuracy of information and satisfaction with supervision may be an artifact of these kinds of systematic response biases.

A third limitation is that scores on communication variables represent only an indication of the adequacy of communication within the team. High-quality audit decisions may be reached irrespective of the degree to which a team possesses the measured variables. A fourth limitation is the use of perceptions of three team members to represent team variables. The experiences of the three participating auditors may not be representative of the experiences of the team. Finally, the representation of audit team structure as a single variable delimits the ability to attribute effects to the components of structure. For instance, we do not know whether audit structure (systematic audit procedures) would yield similar results. The extent to which audit team structure and audit structure overlap is a subject for future research.

APPENDIX

Scales Used in the Study

A. Audit Team Structure (R indicates reverse coding)

On the audit team, most people do not have a voice in decision making.

On the audit team, I am not likely to express my feelings openly about my assignments.

My superior on the audit team usually makes decisions himself/herself.

On the audit team, it is always necessary to go through channels in dealing with decisions concerning my assignments.

Several individuals have a say in making decisions within the audit team (R).

Talking to other people on the audit team about the problems I may have in my work is an important part of the decision-making process (R).

On the audit team, I am encouraged to make suggestions on decisions relevant to my work (R).

On the audit team, my work entails many kinds of activities (R).

I am encouraged to speak my mind on the audit team even if it means disagreeing with my superior (R).

In my work on the audit team, I always have the same areas of responsibility.

On the audit team, if I feel I have the right approach to carrying out my assignment in dealing with a particular problem, I can usually go ahead without checking with my superior (R).

On the audit team, my superior often seeks out my advice before decisions are made (R).

In my work on the audit team, there are many people involved in decision making (R).

On the audit team, the same rules and procedures are generally followed in making decisions.

On the audit team, I have to check with my superior before I do almost anything.

For most situations that arise on the audit team, the rules and procedures are developed as we go along (R).

On the audit team, I do not share any influence with my superior in making decisions.

On the audit team, I always get my orders for my assignments from my superior.

On the audit team, there are rules and procedures for handling most kinds of problems, which may arise in making decisions.

I do not play an active role in making most decisions on the audit team.

On the audit team, my assignments are not clearly defined (R).

B. Information Overload

In a typical week on the audit team, approximately how often do you have *less* than the amount of information you can consistently handle for making the best possible work-related decision?

On the audit team, is the total amount of information you receive in a typical work week *enough* to meet the information requirements of your assignments?

Sometimes on the audit team, you may receive more information than you can efficiently use. At other times, however, you may feel that you are not receiving all the information you need. How often during a week would you say that a *lack* of information arises?

C. Boundary Spanning

How many hours (in an average week) do you attend formal meetings in regard to your assigned work on the audit team:

with people outside of your audit team but within your office?

with people outside of your office altogether?

How many times do you send or receive formal written work-related communication in the form of reports or data to/from others in regard to your assigned work on the audit team:

with people outside of your audit team but within your office?

with people outside of your office altogether?

D. Satisfaction with Supervision

To what extent are you satisfied or dissatisfied with:

the amount of supervision given me on the team?

the guidance by my supervisors on the team in solving work-related problems?

E. Accuracy of Information (each item was reversed coded)

Work-related information communicated to me by other audit team members is often inaccurate.

I can think of a number of times when I received inaccurate work-related information from others on the audit team.

It is often necessary for me to go back and establish the accuracy of work-related information I have received from other audit team members.

I sometimes feel that others on the audit team don't understand the work-related information they have received.

The accuracy of work-related information passed among members of the audit team could be improved.

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